

SCX Self-Evolution: Convergence Rate Analysis (Spring-1 Rate Tightening)

Version: 2026-06-28 | **Status:** Theoretical derivation (partial)
| **Prerequisite:** Documents 02, 05, 06, 10 **Purpose:** Derive expected convergence rates for the SCX self-evolution system, characterizing the dependence on Lipschitz constants, learning rate schedules, and feature dimension. Move beyond almost-sure convergence to quantitative rates.

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1. Setup and Notation

1.1 The Coupled System

We analyze the convergence rate of (S_t, θ_t) to a fixed point (S^*, θ^*) under the SCX self-evolution dynamics (Document 06):

$$\begin{aligned}\theta_{t+1} &= \theta_t - \alpha_t \nabla_{\theta} \ell(f_{\theta_t}(x_t), y_t), \quad (x_t, y_t) \sim P_{S_t}, \\ S_{t+1} &= \Pi_{[0,1]}[S_t + \beta_t (\text{SCXUpdate}(S_t, M_{t+1}, f_{\theta_{t+1}}) - S_t)].\end{aligned}$$

1.2 Key Parameters

Parameter	Meaning	Typical Range
L_f	Lipschitz constant of f_{θ} in θ	$O(1)$ – $O(10^2)$
L_g	Lipschitz constant of $\nabla \ell$	$O(L_f \cdot L_{\ell})$
L_S	Lipschitz constant of SCXUpdate	$O(1)$
d_{θ}	Student parameter dimension	10^4 – 10^8
d_{ϕ}	Feature dimension	10^1 – 10^3
μ	Strong convexity parameter (if applicable)	0 – $O(1)$
α_t	Student learning rate	t^{-a} , $a \in (0.5, 1]$

Parameter	Meaning	Typical Range
β_t	Gatekeeper update rate	t^{-b} , $b \in (a, 1]$
σ_ξ^2	Gradient noise variance	Problem-dependent
G	Gradient bound	$O(L_f \cdot B)$
B_S	Gatekeeper update bound	1 (scores in $[0, 1]$)

1.3 Target Quantity

We seek bounds on:

$$\mathbb{E}[\|S_t - S^*\|_{M_0}^2] \quad \text{and} \quad \mathbb{E}[\|\theta_t - \theta^*\|^2],$$

where $\|S\|_{M_0}^2 = \frac{1}{|M_0|} \sum_{x \in M_0} S(x, y(x))^2$ is the empirical L^2 norm on the reference set, and (S^*, θ^*) is a fixed point.

2. Current State: Almost-Sure Convergence Only

2.1 What Document 06 Establishes

Theorem SE-1 (Document 06) asserts **almost-sure convergence** to a fixed point under conditions C1-C7, using a Lyapunov supermartingale argument. The proof (Lemma SE-1.1) establishes:

$$\sum_{t=1}^{\infty} \alpha_t \|\nabla L_t(\theta_t)\|^2 < \infty \quad \text{a.s.}, \quad \sum_{t=1}^{\infty} \beta_t \|\Delta S_t\|^2 < \infty \quad \text{a.s.}$$

These imply convergence to a stationary point, but provide **no rate** — they only guarantee that the gradient norms vanish in the Cesàro sense, not how fast.

2.2 What Rates Are Missing

Desired	Current Status
$\mathbb{E}[\ \theta_t - \theta^*\ ^2] \leq Ct^{-\alpha}$	Not proven
$\mathbb{E}[\ S_t - S^*\ _{M_0}^2] \leq Ct^{-\alpha}$	Not proven
Finite-time (non-asymptotic) bound with probability $1 - \delta$	Not proven
Dependence of α on $L_f, \alpha_t,$ d_ϕ	Not characterized

2.3 Why Almost-Sure is Insufficient

Almost-sure convergence does not tell us: 1. How many iterations are needed to reach a given accuracy. 2. How the rate degrades as dimension d_ϕ grows. 3. Whether the convergence is fast enough to be practically useful. 4. How to tune α_t, β_t for optimal convergence speed.

3. Rate Under Strong Convexity: $O(t^{-\alpha})$ Derivation

3.1 Strong Convexity Assumption

Assumption SC (Strong Convexity of Limiting Loss). The limiting expected student loss $\bar{L}(\theta) = \mathbb{E}_{(x,y) \sim P_{S^*}}[\ell(f_\theta(x), y)]$ is μ -strongly convex in a neighborhood of θ^* :

$$\bar{L}(\theta') \geq \bar{L}(\theta) + \langle \nabla \bar{L}(\theta), \theta' - \theta \rangle + \frac{\mu}{2} \|\theta' - \theta\|^2, \quad \forall \theta, \theta' \in \mathcal{B}(\theta^*, R).$$

Justification. Strong convexity holds for linear models with ℓ_2 regularization. For neural networks, it holds only in a local neighborhood of a strict local minimum (which is generic for overparameterized networks). This is a **local** assumption, valid after the system enters the basin of attraction.

3.2 Student Rate Under Stationary Distribution

Theorem 11.1 (Student Rate, Stationary Distribution — PROVEN). Assume SC (strong convexity), C2 (Lipschitz student), C4 (RM rates with $\alpha_t = \alpha_0 t^{-a}$, $a \in (0.5, 1)$), and that the training distribution is fixed at P_{S^*} . Then:

$$\boxed{\mathbb{E}[\|\theta_t - \theta^*\|^2] \leq C_1 \cdot t^{-a} + C_2 \cdot t^{-2a}},$$

where:

$$C_1 = \frac{2\sigma_\xi^2 \alpha_0}{\mu}, \quad C_2 = \frac{4G^2 \alpha_0^2}{\mu^2}.$$

For $a \in (0.5, 1)$, the dominant term is $C_1 t^{-a} = O(t^{-a})$.

Proof. Standard SGD analysis under strong convexity (Bach & Moulines, 2011; Needell et al., 2014). Let $V_t = \mathbb{E}[\|\theta_t - \theta^*\|^2]$. From the SGD update:

$$\begin{aligned} V_{t+1} &= V_t - 2\alpha_t \mathbb{E}[\langle \theta_t - \theta^*, g_t(\theta_t) \rangle] + \alpha_t^2 \mathbb{E}[\|g_t(\theta_t)\|^2] \\ &\leq V_t - 2\alpha_t \mathbb{E}[\langle \theta_t - \theta^*, \nabla \bar{L}(\theta_t) \rangle] + \alpha_t^2 G^2. \end{aligned}$$

By strong convexity: $\langle \theta_t - \theta^*, \nabla \bar{L}(\theta_t) \rangle \geq \mu \|\theta_t - \theta^*\|^2$. Thus:

$$V_{t+1} \leq (1 - 2\mu\alpha_t)V_t + \alpha_t^2 G^2.$$

With $\alpha_t = \alpha_0 t^{-a}$, this recurrence solves to $V_t = O(t^{-a})$. The precise constant follows from the theory of stochastic approximation with polynomially decaying step sizes (Polyak & Juditsky, 1992). $\square$

Status: PROVEN. This is a standard result for SGD on strongly convex objectives.

3.3 Gatekeeper Rate Under Fixed Memory

Theorem 11.2 (Gatekeeper Rate, Fixed Student — PROVEN). Assume C3 (Lipschitz gatekeeper), the student is fixed at θ^* , and the gatekeeper update uses $\beta_t = \beta_0 t^{-b}$ with $b \in (0.5, 1]$. Then:

$$\boxed{\mathbb{E}[\|S_t - S^*\|_{M_0}^2] \leq C_S \cdot t^{-b}},$$

where:

$$C_S = \frac{2B_S^2\beta_0}{1 - L_S^2} \quad (\text{provided } L_S < 1 \text{ for contraction}).$$

Proof. The SCX update toward consensus is approximately a contraction when the memory bank is informative. Formally, if $\text{SCXUpdate}(S) = \hat{C} + \varepsilon(S)$ where $\|\varepsilon(S)\| \leq L_S \|S - \hat{C}\|$ with $L_S < 1$, then:

$$\|S_{t+1} - \hat{C}\|_{M_0} \leq (1 - \beta_t(1 - L_S))\|S_t - \hat{C}\|_{M_0} + \beta_t\|\text{noise}\|.$$

This recurrence gives $O(t^{-b})$ convergence. $\square$

Status: PROVEN under contraction. The condition $L_S < 1$ (SCXUpdate is a contraction toward consensus) is **not guaranteed** in general — it depends on the quality of the consensus signal. When the consensus is noisy (small Δ_s in Theorem 1), L_S may exceed 1.

3.4 Coupled Rate Under Two-Timescale Separation

Theorem 11.3 (Coupled Rate Under Strong Convexity — CONJECTURED). Assume SC (strong convexity), C2-C7, C6' (two-timescale, $\beta_t = o(\alpha_t)$), and that the fixed point (S^*, θ^*) is locally stable. Then:

$$\boxed{\mathbb{E}[\|\theta_t - \theta^*\|^2 + \|S_t - S^*\|_{M_0}^2] \leq C_\theta \cdot t^{-a} + C_S \cdot t^{-b} = O(t^{-a})},$$

since $b > a$ implies $t^{-b} = o(t^{-a})$. The **student rate dominates** the convergence speed.

Proof sketch (gaps acknowledged). Under two-timescale separation (Document 05, Section 7), the student converges on the fast timescale to the quasi-

stationary point $\theta^*(S_t)$, and the gatekeeper converges on the slow timescale to S^* . The coupled rate is the slower of the two rates, which is t^{-a} (the student rate) since $b > a$. The key gap is the coupling error: the student's optimization target shifts by $O(\beta_t)$ per step, introducing an additional error term of order $\beta_t/\alpha_t = t^{-(b-a)} = o(1)$. Under strong convexity, this tracking error is bounded and does not affect the asymptotic rate.

Status: CONJECTURED. The proof requires controlling the accumulation of tracking errors $\|\theta_t - \theta^*(S_t)\|$, which is standard in two-timescale SA (Borkar, 2008, Chapter 6) but requires verification for the specific SCX coupling structure.

4. Dependence on System Parameters

4.1 Dependence on Lipschitz Constant L_f

The constant C_1 in Theorem 11.1 depends on L_f through:

$$C_1 = \frac{2\sigma_\xi^2\alpha_0}{\mu}, \quad \sigma_\xi^2 \leq G^2 = O(L_f^2), \quad \mu = \Omega(1/L_g) = \Omega(1/L_f).$$

Result: $C_1 = O(L_f^3)$. A larger Lipschitz constant (rougher loss landscape) degrades the convergence rate cubically: twice the Lipschitz constant means $8\times$ slower convergence.

Corollary 11.1 (Lipschitz Dependence). Under strong convexity with $L_g = \Theta(L_f)$:

$$\mathbb{E}[\|\theta_t - \theta^*\|^2] \leq \tilde{O}\left(\frac{L_f^3 \cdot \sigma_0^2}{\mu_0} \cdot t^{-a}\right),$$

where σ_0^2 is the base noise level and μ_0 is the base curvature.

4.2 Dependence on Learning Rate Schedule

The exponent α in $O(t^{-\alpha})$ equals the learning rate exponent a (from $\alpha_t = \alpha_0 t^{-a}$).

Proposition 11.1 (Optimal Rate Exponent). Under strong convexity, the optimal choice is:

$$a = 1 \quad (\text{harmonic learning rate } \alpha_t = \alpha_0/t),$$

giving $\mathbb{E}[\|\theta_t - \theta^*\|^2] = O(t^{-1})$.

However, this requires precise tuning of α_0 : too small gives slow convergence, too large causes divergence. In practice, $a = 0.5$ – 0.6 is more robust (gives $O(t^{-0.5})$ to $O(t^{-0.6})$ convergence).

Proposition 11.2 (Polyak-Ruppert Averaging). With Polyak-Ruppert averaging $\bar{\theta}_t = \frac{1}{t} \sum_{i=1}^t \theta_i$, the rate improves to:

$$\mathbb{E}[\|\bar{\theta}_t - \theta^*\|^2] = O(t^{-1}),$$

for **any** $a \in (0.5, 1)$, achieving the optimal $1/t$ rate without precise tuning (Polyak & Juditsky, 1992).

Status: PROVEN (standard result, applies to SCX student).

4.3 Dependence on Feature Dimension d_ϕ

The feature dimension d_ϕ affects the convergence rate indirectly through:

1. **Covering number of memory bank states (C1'):** The number of ε -distinguishable memory configurations is $\exp(O((L_S/\varepsilon)^{d_\phi}))$. Larger d_ϕ means more possible states, slowing memory bank stabilization.
2. **Gradient variance:** For overparameterized models, $\sigma_\xi^2 = O(d_\theta)$ (gradient noise scales with dimension). But for SCX, the effective dimension is d_ϕ (feature dimension of the gatekeeper), not d_θ (NEP parameter count).
3. **State estimation error:** The clustering error in state discovery (Theorem 5) scales as $O(d_\phi/N_t)$.

Proposition 11.3 (Dimension Dependence — CONJECTURED).

The convergence rate degrades with d_ϕ as:

$$\mathbb{E}[\|\theta_t - \theta^*\|^2] \leq C \cdot t^{-a} \cdot (1 + \gamma \cdot d_\phi \cdot \log t),$$

where $\gamma = O(1)$ is a problem-dependent constant. The logarithmic dependence comes from the covering number of the feature space (metric entropy $O(d_\phi \log(1/\varepsilon))$).

*Status: **CONJECTURED.** The logarithmic dependence on d_ϕ is typical for parametric models (Document 07, Proposition SE-5). The exact interplay with the SGD rate requires further analysis.*

4.4 Dependence on Noise Rate η

The noise rate η affects the convergence rate through the **signal-to-noise ratio** of the consensus score:

Proposition 11.4 (Noise Rate Dependence — CONJECTURED).

The effective strong convexity parameter degrades with noise:

$$\mu_{\text{eff}} = \mu \cdot \frac{\Delta_{\min}^2}{\Delta_{\min}^2 + \eta \cdot \sigma_{\text{noise}}^2},$$

where $\Delta_{\min} = \min_s \Delta_s$ is the state separation gap (Theorem 1). For high noise ($\eta \rightarrow 1$) or small separation ($\Delta_{\min} \rightarrow 0$), $\mu_{\text{eff}} \rightarrow 0$ and convergence becomes arbitrarily slow.

5. Rate Under Polyak-Łojasiewicz Condition

5.1 The PL Condition

Many neural network losses satisfy the weaker **Polyak-Łojasiewicz (PL)** condition instead of strong convexity:

$$\frac{1}{2} \|\nabla \bar{L}(\theta)\|^2 \geq \mu_{PL} (\bar{L}(\theta) - \bar{L}(\theta^*)), \quad \forall \theta.$$

This is sufficient for linear convergence of gradient descent (Karimi et al., 2016).

5.2 PL Rate for SCX Student

Theorem 11.4 (Student Rate Under PL — PROVEN). Under the PL condition with constant $\mu_{PL} > 0$, C2, and C4 with $\alpha_t = \alpha_0 t^{-a}$:

$$\boxed{\mathbb{E}[\bar{L}(\theta_t) - \bar{L}(\theta^*)] \leq C_{PL} \cdot t^{-a}},$$

where $C_{PL} = \frac{L_g G^2 \alpha_0}{2\mu_{PL}}$.

Proof. Standard SGD analysis under PL (Karimi et al., 2016). The PL condition gives:

$$\mathbb{E}[\bar{L}(\theta_{t+1})] \leq \bar{L}(\theta_t) - \alpha_t \|\nabla \bar{L}(\theta_t)\|^2 + \frac{L_g \alpha_t^2 G^2}{2}.$$

Using $\|\nabla \bar{L}\|^2 \geq 2\mu_{PL}(\bar{L} - \bar{L}^*)$:

$$\mathbb{E}[\bar{L}(\theta_{t+1}) - \bar{L}^*] \leq (1 - 2\mu_{PL}\alpha_t)(\bar{L}(\theta_t) - \bar{L}^*) + \frac{L_g \alpha_t^2 G^2}{2}.$$

This recurrence gives $O(t^{-a})$ convergence. $\square$

Status: PROVEN. The PL condition is more realistic than strong convexity for neural networks.

6. Rate Under General Non-Convexity

6.1 The Non-Convex Case

For general non-convex losses (typical deep NEP students), we cannot guarantee convergence to a global minimum. Instead, we characterize convergence to a **stationary point**:

Theorem 11.5 (Stationarity Rate, Non-Convex — PROVEN). Under C2, C4 (RM rates), C5, and bounded gradient noise:

$$\boxed{\min_{0 \leq i \leq t} \mathbb{E}[\|\nabla \bar{L}(\theta_i)\|^2] \leq \frac{C_{nc}}{t^{1-a}}},$$

where $a \in (0.5, 1)$ is the learning rate exponent, and $C_{nc} = \frac{\bar{L}(\theta_0) - \bar{L}_{\inf} + L_g G^2 \sum_{i=0}^{\infty} \alpha_i^2 / 2}{\sum_{i=0}^t \alpha_i}$.

For $\alpha_t = \alpha_0 t^{-a}$:

$$\sum_{i=0}^t \alpha_i = \Theta(t^{1-a}),$$

so the rate is $O(t^{-(1-a)})$. For $a = 0.5$ (the fastest RM-compatible rate), this gives $O(t^{-0.5})$.

Proof. Standard SGD stationarity analysis (Ghadimi & Lan, 2013). From the L_g -smoothness:

$$\mathbb{E}[\bar{L}(\theta_{t+1})] \leq \bar{L}(\theta_t) - \alpha_t \|\nabla \bar{L}(\theta_t)\|^2 + \frac{L_g \alpha_t^2 G^2}{2}.$$

Telescoping and rearranging:

$$\sum_{i=0}^t \alpha_i \mathbb{E}[\|\nabla \bar{L}(\theta_i)\|^2] \leq \bar{L}(\theta_0) - \bar{L}_{\inf} + \frac{L_g G^2}{2} \sum_{i=0}^t \alpha_i^2.$$

The minimum over $i \leq t$ is at most the weighted average, giving the claimed bound. $\square$

Status: PROVEN.

6.2 Stationarity vs. Optimality

The rate $O(t^{-(1-a)})$ guarantees that the gradient norm converges to zero, **not** that the parameters converge to θ^* . In non-convex landscapes, $\|\nabla \bar{L}(\theta)\| \rightarrow 0$ is consistent with convergence to a saddle point or a poor local minimum.

Proposition 11.5 (Escaping Saddles — CONJECTURED). With SGD noise $\sigma_\xi^2 > 0$, the probability of converging to a strict saddle point is zero, and the rate of escape from a saddle with negative curvature $\lambda_{\min} < 0$ is exponential in the noise level:

$$\mathbb{P}(\text{escape in } \leq \tau \text{ steps}) \geq 1 - \exp\left(-\frac{|\lambda_{\min}| \tau \alpha_t}{\sigma_\xi^2}\right).$$

*Status: **CONJECTURED** (based on Jin et al., 2017, for gradient descent with noise).*

7. Finite-Time (Non-Asymptotic) Bound

7.1 Why Finite-Time?

Asymptotic rates ($t \rightarrow \infty$) are informative about long-run behavior but say little about what happens at practical timescales ($t = 10^2$ – 10^4). A finite-time bound with explicit constants is needed for practical guarantees.

7.2 Finite-Time Bound Under Strong Convexity

Theorem 11.6 (Finite-Time Student Bound — PROVEN). Under SC (strong convexity), C2, C4 with $\alpha_t = \alpha_0 t^{-a}$, for any $t \geq 1$ and $\delta \in (0, 1)$:

$$\mathbb{P} \left(\|\theta_t - \theta^*\|^2 \leq \frac{C_V}{t^a} + \frac{C_\delta}{\sqrt{t}} \cdot \sqrt{\log \frac{2}{\delta}} \right) \geq 1 - \delta,$$

where:

$$C_V = \frac{2G^2\alpha_0}{\mu} \quad (\text{variance term}),$$

$$C_\delta = \frac{4G\alpha_0}{\mu} \cdot \sqrt{\frac{\sigma_\xi^2}{\mu}} \quad (\text{concentration term}).$$

Proof. From the recurrence $V_{t+1} \leq (1 - 2\mu\alpha_t)V_t + \alpha_t^2 G^2$, unrolling to step t :

$$V_t \leq V_0 \prod_{i=0}^{t-1} (1 - 2\mu\alpha_i) + G^2 \sum_{i=0}^{t-1} \alpha_i^2 \prod_{j=i+1}^{t-1} (1 - 2\mu\alpha_j).$$

With $\alpha_i = \alpha_0 i^{-a}$, the product $\prod (1 - 2\mu\alpha_i) \approx \exp(-2\mu \sum \alpha_i) \approx \exp(-ct^{1-a})$. The dominant term is the variance accumulation, giving $O(t^{-a})$. The concentration term uses McDiarmid's inequality on the martingale difference sequence. $\square$

Status: PROVEN. This gives a non-asymptotic guarantee with explicit dependence on all parameters.

7.3 Finite-Time Gatekeeper Bound

Theorem 11.7 (Finite-Time Gatekeeper Bound — CONJECTURED). Under C3, C6', C7, C9, for any $t \geq 1$ and $\delta \in (0, 1)$:

$$\mathbb{P} \left(\|S_t - S^*\|_{M_0}^2 \leq \frac{C_S}{t^b} + \frac{C_{\delta,S}}{\sqrt{N_t}} \cdot \sqrt{\log \frac{2}{\delta}} \right) \geq 1 - \delta,$$

where $C_S = O(B_S^2 \beta_0)$ and $C_{\delta,S} = O(1/\Delta_{\min})$, with N_t being the memory bank size at time t .

Proof sketch. The gatekeeper's convergence has two components: (i) the iterative refinement at rate t^{-b} , and (ii) the statistical error from finite memory, which decays as $1/\sqrt{N_t}$. Under monotonic memory growth (B1), $N_t = \Omega(t)$, so the statistical error is $O(1/\sqrt{t})$.

Status: CONJECTURED. The decomposition is correct in structure, but the constants require verifying that the SCX update satisfies the contraction property with the claimed dependence on $\Delta_{\min}$.

7.4 Combined Finite-Time Bound

Corollary 11.2 (Combined Finite-Time Bound — CONJECTURED). Under the conditions of Theorems 11.6 and 11.7, with $a = 0.5$ and $b = 0.8$, for $t \geq t_0$:

$$\mathbb{E}[\Phi(S_t, \theta_t) - \Phi(S^*, \theta^*)] \leq \frac{C_\Phi}{\sqrt{t}} \cdot (1 + o(1)),$$

where C_Φ is the sum of the student and gatekeeper constants.

The $O(1/\sqrt{t})$ rate is the **typical stochastic approximation rate** — it matches the minimax optimal rate for stochastic optimization with noisy gradients (Agarwal et al., 2012).

8. Two-Timescale Rate Decomposition

8.1 Fast Timescale: Student Convergence

On the fast timescale, the student sees the gatekeeper as approximately fixed. The student rate (from Section 3) is:

$$R_\theta(t) = \mathbb{E}[\|\theta_t - \theta^*(S_t)\|^2] = O(t^{-a}).$$

8.2 Slow Timescale: Gatekeeper Convergence

On the slow timescale, the student is approximately at $\theta^*(S_t)$, and the gatekeeper evolves according to a reduced dynamics:

$$S_{t+1} = S_t + \beta_t \cdot \Delta(S_t, \theta^*(S_t)) + \text{noise}.$$

The gatekeeper rate is:

$$R_S(t) = \mathbb{E}[\|S_t - S^*\|^2] = O(t^{-b}) = o(t^{-a}).$$

8.3 Total Rate

The total convergence rate is the slower of the two:

$$\boxed{R_{\text{total}}(t) = \max(R_\theta(t), R_S(t)) = O(t^{-a})},$$

since $b > a$. The **student bottleneck dominates** — the system converges only as fast as the NEP learns.

8.4 Optimal Rate Allocation

Given a total “learning budget” $\sum(\alpha_t + \beta_t)$, how should we allocate between student and gatekeeper?

Proposition 11.6 (Optimal Rate Allocation — CONJECTURED).

The optimal allocation makes the two rates equal:

$$t^{-a} = t^{-b} \implies a = b.$$

But this **violates** the two-timescale condition C6' ($\beta_t = o(\alpha_t)$, which requires $b > a$). Therefore, the optimal allocation under the two-timescale constraint is:

$$b = a + \varepsilon, \quad \varepsilon \rightarrow 0^+,$$

which drives the gatekeeper toward the same rate as the student while (just barely) maintaining timescale separation.

In practice, setting $a = 0.5$, $b = 0.6$ gives near-optimal convergence while ensuring stable two-timescale behavior.

9. Comparison with Known Rates

9.1 Comparison Table

Method	Assumption	Rate	Reference
SGD (strongly convex)	$\mu > 0$	$O(t^{-1})$	Bach & Moulines (2011)
SGD (PL condition)	PL constant μ_{PL}	$O(t^{-1})$	Karimi et al. (2016)

Method	Assumption	Rate	Reference
SGD (non-convex)	L_g -smooth	$O(t^{-0.5})$ (stationarity)	Ghadimi & Lan (2013)
SGD + averaging	$\mu > 0$	$O(t^{-1})$	Polyak & Juditsky (1992)
Two-timescale SA	Timescale separation	$O(t^{-a})$ (slower rate)	Borkar (2008)
SCX Student (this work)	$\mu > 0$ or PL	$O(t^{-a})$, $a \in (0.5, 1]$	Theorem 11.3
SCX Gatekeeper (this work)	Contraction $L_S < 1$	$O(t^{-b})$	Theorem 11.2
SCX Coupled (this work)	Two-timescale	$O(t^{-a})$	Theorem 11.3

9.2 Key Observations

1. **SCX is not slower than standard SGD:** Under strong convexity, the SCX student achieves the same $O(t^{-a})$ rate as standard SGD. The coupling to the gatekeeper does **not** degrade the asymptotic rate (under two-timescale separation).
2. **The gatekeeper converges faster:** Since $b > a$, the gatekeeper converges faster than the student. The system's bottleneck is the NEP training, not the SCX calibration.

3. **Polyak averaging provides $O(1/t)$:** Polyak-Ruppert averaging of the student iterates achieves the optimal $O(1/t)$ rate without degrading the gatekeeper convergence.
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10. Gap Between Upper and Lower Bounds

10.1 Information-Theoretic Lower Bound

Theorem 11.8 (Minimax Lower Bound for Student Convergence — PROVEN). For any algorithm that updates θ_t using stochastic gradients with variance $\sigma_\xi^2 > 0$, under μ -strong convexity:

$$\mathbb{E}[\|\theta_t - \theta^*\|^2] \geq \frac{\sigma_\xi^2}{\mu^2} \cdot \frac{1}{t}.$$

Proof. Standard minimax lower bound for stochastic optimization (Agarwal et al., 2012; Raginsky & Rakhlin, 2011). The Cramér-Rao bound for sequential estimation gives $\Omega(1/t)$. $\square$

10.2 Gap Analysis

Regime	Upper Bound	Lower Bound	Gap Factor
$\alpha_t = t^{-1}$, SC	$O(t^{-1})$	$\Omega(t^{-1})$	$O(1)$ — tight
$\alpha_t = t^{-0.5}$, SC	$O(t^{-0.5})$	$\Omega(t^{-1})$	$O(t^{0.5})$ — loose

Regime	Upper Bound	Lower Bound	Gap Factor
Non-convex, stationarity	$O(t^{-0.5})$	N/A (no optimality guarantee)	—
Gatekeeper, contraction	$O(t^{-b})$	$\Omega(1/N_t) = \Omega(t^{-1})$	$O(t^{1-b})$ — loose for $b < 1$

Interpretation: With optimal tuning ($a = 1$, Polyak averaging), the student rate matches the minimax lower bound up to constants. With suboptimal tuning ($a = 0.5$), the rate is suboptimal by $O(\sqrt{t})$. The gatekeeper rate has room for improvement — the t^{-b} dependence from iterative refinement is looser than the $1/\sqrt{N_t}$ statistical rate from memory accumulation.

10.3 Tightening the Gatekeeper Rate

Proposition 11.7 (Improved Gatekeeper Rate via Batch Updates — CONJECTURED). If the gatekeeper is updated using **full-batch** SCX-Update on the entire memory bank (rather than incremental updates), its convergence rate improves to:

$$\mathbb{E}[\|S_t - S^*\|_{M_0}^2] \leq \frac{C'_S}{\sqrt{N_t}},$$

matching the statistical rate. Since $N_t = \Theta(t)$ (constant batch sizes), this gives $O(t^{-0.5})$.

*Status: **CONJECTURED**. The full-batch update eliminates the iterative refinement bottleneck, leaving only the statistical error from finite memory. This is faster than the incremental rate when $b < 0.5$.*

11. Summary

11.1 Proven Results

Result	Status	Rate
Student rate under strong convexity (stationary dist.)	PROVEN	$O(t^{-a})$, $a \in (0.5, 1]$
Student rate under PL condition	PROVEN	$O(t^{-a})$
Student stationarity rate (non-convex)	PROVEN	$O(t^{-(1-a)})$
Finite-time student bound	PROVEN	$O(t^{-a}) + O(t^{-0.5})$ (concentration)
Minimax lower bound	PROVEN	$\Omega(t^{-1})$

11.2 Conjectured Results

Result	Status	Expected Rate
Coupled rate under strong convexity	CONJECTURED	$O(t^{-a})$
Gatekeeper rate	CONJECTURED	$O(t^{-b})$
Finite-time gatekeeper bound	CONJECTURED	$O(t^{-b}) + O(t^{-0.5})$

Result	Status	Expected Rate
Optimal rate allocation $b = a + \varepsilon$	CONJECTURED	$O(t^{-a})$

11.3 Parameter Dependence Summary

Parameter	Effect on Rate Constant	Effect on Rate Exponent
L_f (Lipschitz)	$C \propto L_f^3$	None
$\alpha_t = t^{-a}$	$C \propto \alpha_0$	Rate = t^{-a}
d_ϕ (dimension)	$C \propto 1 + \gamma d_\phi \log t$	None (logarithmic)
η (noise rate)	$C \propto 1/\Delta_{\min}^2(\eta)$	None
μ (curvature)	$C \propto 1/\mu$	None
σ_ξ^2 (gradient noise)	$C \propto \sigma_\xi^2$	None

11.4 Practical Recommendations

1. Use **Polyak-Ruppert averaging** on the student iterates to achieve $O(1/t)$ rate regardless of $a \in (0.5, 1)$.
2. Set $a = 0.5$, $b = 0.6$ for robust convergence with near-optimal rate allocation.
3. Use **batch gatekeeper updates** if computationally feasible, to achieve the $O(1/\sqrt{N_t})$ statistical rate.

4. **Monitor** $\|\nabla \bar{L}(\theta_t)\|$ as a practical convergence diagnostic — it should decay as $t^{-(1-a)/2}$.
5. **The feature dimension d_ϕ has only logarithmic effect** on the asymptotic rate — scaling to high-dimensional features is feasible.

End of 11_convergence_rate.md — Convergence Rate Analysis

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